Li Ma

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Biography

Li Ma is currently a Senior Research Scientist at Netflix Eyeline Labs, working with Paul Debevec and Eric Roth. He graduated with a Ph.D. from the Department of Computer Science & Engineering at the Hong Kong University of Science and Technology (HKUST). His Ph.D. supervisor is Prof. Pedro V. Sander, and he works closely with Prof. Liao Jing. Before that, he obtained his Bachelor's degree with honors in Electrical Engineering from Zhejiang University in 2019.
His current research focuses on generative AI techniques for content creation and film production, with a particular emphasis on controllable image and video generation. During PhD, he specialized in 3D/4D reconstruction, novel view synthesis, neural rendering, differentiable rendering, and other graphics-related vision topics.

Work Experience

Netflix Eyeline Labs — **Los Angeles, US**
Senior Research Scientist — *Jan 2026 – Now*

- Manager: Paul Debevec
- GenAI for film production

Netflix Eyeline Labs — **Los Angeles, US**
Research Scientist — *April 2024 – Dec 2025*

- Manager: Paul Debevec
- GenAI for film production

Meta Reality Labs — **Pittsburgh, US**
Research Scientist Intern — *May. 2023 – Oct. 2023*

- Manager: Michael Zollhoefer and Christian Richardt
- Modeling reflections in NeRF and inverse rendering
- Reconstruct NeRF for VR rendering

Tencent AI Lab — **Shenzhen, China**
Research Intern — *Aug. 2021 – Sept. 2022*

- Manager: Xiaoyu Li
- NeRF reconstruction under non-ideal input
- 3D Digital human face modeling and editing

Education

The Hong Kong University of Science and Technology — **Hong Kong, China**
Ph.D., Computer Science & Engineering — *Sept. 2019 – Jan. 2024*
Advised by: Prof. Pedro V. Sander

Zhejiang University — **Zhejiang, China**
B.Eng. Electrical Engineering — *Sept. 2015 – Jun. 2019*
GPA: 3.95/4.0; RANK: 2/89

Publications

* indicates joint first author

- BodyReLux: Temporally Consistent Whole-body Relighting
  Li Ma, Mingming He, Xueming Yu, David M George, Ahmet Levent Taşel, Ning Yu, Paul Debevec, Julien Philip
  Siggraph 2026, Journal Track
- Go-with-the-Track: Video Compositing and Motion Control with Point Tracking
  Koichi Namekata, Yash Kant, Zhizheng Liu, Ryan D Burgert, Kuan Heng Lin, Yuancheng Xu, Emmett Steven, Pablo Salamanca, Julien Philip, **Li Ma**, Andrea Vedaldi, Paul Debevec, Ning Yu
  Siggraph 2026
- DiffHDR: Re-Exposing LDR Videos with Video Diffusion Models
  Zhengming Yu, **Li Ma**, Mingming He, Leo Isikdogan, Yuancheng Xu, Dmitriy Smirnov, Pablo Salamanca, Dao Mi, Pablo Delgado, Ning Yu, Julien Philip, Xin Li, Wenping Wang, Paul Debevec
  Under Review, 2026
- ID-V2V: Identity-Preserving Video Restylization
  Yuancheng Xu, Mingming He, Pablo Salamanca, **Li Ma**, Yash Kant, Emmett Steven, Paul Debevec, Ning Yu
  Under Review, 2026
- RetimeGS: Continuous-Time Reconstruction of 4D Gaussian Splatting
  Xuezhen Wang, **Li Ma**, Yulin Shen, Zeyu Wang, Pedro V. Sander
  CVPR 2026, Oral
- Lighting in Motion: Spatiotemporal HDR Lighting Estimation
  Christophe Bolduc, Julien Philip, **Li Ma**, Mingming He, Paul Debevec, Jean-François Lalonde
  CVPR 2026
- Detail Enhanced Gaussian Splatting for Large-Scale Volumetric Capture
  Julien Philip*, **Li Ma***, Pascal Clausen*, Wenqi Xian*, Ahmet Levent Taşel, Mingming He, Xueming Yu, David M George, Ning Yu, Oliver Pilarski, Paul Debevec
  Siggraph Asia 2025, Journal Track | *Project Page*
- GSWT: Gaussian Splatting Wang Tiles
  Yunfan Zeng, **Li Ma**, Pedro V. Sander
  Siggraph Asia 2025 | *Project Page*
- Virtually Being: Customizing Camera-Controllable Video Diffusion Models with Multi-View Performance Captures
  Yuancheng Xu, Wenqi Xian, **Li Ma**, Julien Philip, Ahmet Levent Taşel, Yiwei Zhao, Ryan Burgert, Mingming He, Oliver Hermann, Oliver Pilarski, Rahul Garg, Paul Debevec, Ning Yu
  Siggraph Asia 2025 | *Project Page*
- Go-with-the-Flow: Motion-Controllable Video Diffusion Models Using Real-Time Warped Noise
  Ryan Burgert, Yuancheng Xu, Wenqi Xian, Oliver Pilarski, Pascal Clausen, Mingming He, **Li Ma**, Yitong Deng, Lingxiao Li, Mohsen Mousavi, Michael Ryoo, Paul Debevec, Ning Yu
  CVPR 2025, Oral | *Project Page*
- Lux Post Facto: Learning Portrait Performance Relighting with Conditional Video Diffusion and a Hybrid Dataset
  Yiqun Mei, Mingming He, **Li Ma**, Julien Philip, Wenqi Xian, David George, Xueming Yu, Gabriel Dedic, Ahmet Levent Taşel, Ning Yu, Vishal M. Patel, Paul Debevec

CVPR 2025 | *Project Page*

- DifFRelight: Diffusion-Based Facial Performance Relighting
  Mingming He*, Pascal Clausen*, Ahmet Levent Taşel*, **Li Ma***, Oliver Pilarski*, Wenqi Xian, Laszlo Rikker, Xueming Yu, Ryan Burgert, Ning Yu, Paul Debevec
  Siggraph Asia 2024 | *Project Page*
- Fitting Spherical Gaussians to Dynamic HDRI Sequences
  Pascal Clausen*, **Li Ma***, Mingming He*, Ahmet Levent Taşel*, Oliver Pilarski*, Paul Debevec
  Siggraph Asia Posters 2024 | *Project Page*
- Survey of Video Diffusion Models: Foundations, Implementations, and Applications
  Yimu Wang*, Xuye Liu*, Wei Pang*, **Li Ma***, Shuai Yuan*, Paul Debevec, Ning Yu
  TMLR 2025 | *Arxiv*
- Fisheye-GS: Lightweight and Extensible Gaussian Splatting Module for Fisheye Cameras
  Zimu Liao, Siyan Chen, Rong Fu, Yi Wang, Zhongling Su, Hao Luo, **Li Ma**, Linning Xu, Bo Dai, Hengjie Li, Zhilin Pei, Xingcheng Zhan
  ECCV Workshop 2024 | *Arxiv*
- SpecNeRF: Gaussian Directional Encoding for Specular Reflections
  Li Ma, Vasu Agrawal, Haithem Turki, Changil Kim, Chen Gao, Pedro V. Sander, Michael Zollhöfer, Christian Richardt
  CVPR 2024, Highlight | *Project Page*
- 3D Video Loops from Asynchronous Input
  Li Ma, Xiaoyu Li, Jing Liao, Pedro V. Sander
  CVPR 2023 | *Project Page*
- Neural Parameterization for Dynamic Human Head Editing
  Li Ma, Xiaoyu Li, Jing Liao, Xuan Wang, Qi Zhang, Jue Wang, Pedro V. Sander
  Siggraph Asia 2022 | *Project Page*
- Deblur-NeRF: Neural Radiance Fields from Blurry Images
  Li Ma, Xiaoyu Li, Jing Liao, Qi Zhang, Xuan Wang, Jue Wang, Pedro V. Sander
  CVPR 2022 | *Project Page*
- TVConv: Efficient Translation Variant Convolution
  Jierun Chen, Tianlang He, Weipeng Zhuo, **Li Ma**, Sangtae Ha, S.-H. Gary Chan
  CVPR 2022
- VI-Eye: Semantic-based 3D Point Cloud Registration for Infrastructure-assisted Autonomous Driving
  Yuze He, **Li Ma**, Zhehao Jiang, Yi Tang, Guoliang Xing
  MobiCom 2021

Selected Honors & Awards

- Postgraduate Scholarship, HKUST, 2019 - 2023
- The First Prize (Rank: 2/67), The NXP Cup Undergraduate Intelligent Car Race of Zhejiang Province, 2018
- The Second Prize, The NXP Cup Undergraduate Intelligent Car Race of China, 2018
- Research and Innovation Scholarship, Zhejiang University, 2018
- UHV Scholarship, China, 2018
- First Class Scholarship for Outstanding Merits, Zhejiang University, 2017

- Excellent Student Awards, Zhejiang University, 2016

Teaching Assistant

- COMP5411 Advanced Computer Graphics, *2022 Fall* & *2020 Fall*
- COMP2011 Programming with C++, *2020 Spring*